

Fernando VAN DER VINNE

Mechanical Design Engineer | Robotics & Product Development

[in linkedin.com/in/fernando-vinne](https://www.linkedin.com/in/fernando-vinne)  Munich, Germany

 Portfolio

 EU citizen



Mechanical Design Engineer with hands-on experience in robotics, product development, CAD design, prototyping, and mechanical systems integration. Strong background in developing robot components, end-effectors, 3D-printed parts, and functional prototypes from concept to testing. Passionate about creating practical engineering solutions at the intersection of mechanical design, automation, and robotics.

PROFESSIONAL EXPERIENCE & ENGINEERING PROJECTS

Present Sep 2024	Mechanical Design Engineer , sewts GmbH, Munich <i>Full-time Intern: Sep 2024–Sep 2025 Working Student: Sep 2025–Present</i> ➤ Served as primary mechanical designer for key hardware modules of a robotic cell for automated textile handling in industrial laundries ➤ Developed a functional textile gripper concept that addressed a long-standing technical challenge in automated laundry handling and was used in 2 deployed gripper prototypes ➤ Designed and tested 5+ gripper concepts and 20+ rapid-prototyped or silicone-cast iterations to improve textile pickup performance under realistic laundry conditions ➤ Contributed to 2 deployed versions of a dirty-laundry separator system, from early concept to industrial deployment, with a 3rd version currently in development ➤ Designed robot end-effectors, silicone fingers, gripper structures, pneumatic concepts, robot cabling, fixtures, transport and assembly concepts for industrial use ➤ Developed CAD assemblies, molds, 3D-printed parts, mechanical test setups, and hardware used for AI training data collection in customer laundry environments Autodesk Inventor End-Effectors Product development Pneumatics Silicone Casting Prototyping Industrial Automation CNC Sheet metal FDM
Aug 2024 Sep 2021	Research Assistant - Mechanical Design & Robotics, TUM LPL, Garching ➤ Designed and developed mechanical components for robotic systems, including joints, structural parts, and test assemblies ➤ Created and optimized CAD models for research prototypes, manufacturing, assembly, and finite element analysis ➤ Supported the development of robotic arm modules, with a focus on mechanical stability, load transfer, bearing integration, and modularity ➤ Managed and maintained FDM and SLA 3D printers used for prototyping and research projects Fusion 360 Product development FDM SLA
May 2024 Dec 2023	Bachelor Thesis – Modular Low-Cost Robot Arm, Grade: 1.3, TUM LPL, Garching ➤ Developed modular mechanical concepts for low-cost robotic arm systems, improving stability, load handling, and assembly accessibility ➤ Redesigned key robot arm modules through improved bearing selection, load distribution, and mechanical integration ➤ Addressed alignment, compatibility, and maintenance challenges in a modular robotic system ➤ Designed end-effectors for specific laboratory automation tasks Robot Arm Design Modular Design Bearing selection Mechanical Integration End-Effectors CNC FDM
Feb 2023 Oct 2022	Project Seminar - Exoskeleton Knee Joint, TUM LPL, Garching ➤ Developed a knee joint concept for an exoskeleton supporting patients with limited movement capacity ➤ Contributed to the design of a series elastic actuator, combining mechanical design, electronics, control, simulation, and prototyping Exoskeleton Series Elastic Actuator Mechatronics Prototyping Product Development

Sep 2019	Manufacturing Intern, REINDL MASCHINENBAU GMBH, Glonn > Gained hands-on experience in manufacturing, assembly, quality control, and CNC-based production processes > Supported the production and installation of mechanical equipment, mainly for automotive manufacturing applications
Aug 2019	

Manufacturing
Assembly
Quality Control
CNC
Automotive Production

EDUCATION

Dec 2027	M.Sc. Mechanical Engineering - Technical University of Munich (<i>expected</i>)
2019–2025	B.Sc. Mechanical Engineering - Technical University of Munich
2018–2019	Studienkolleg T-Kurs - TU Dresden, Institute of Advanced Studies

TECHNICAL SKILLS

CAD & Mechanical Design	Autodesk Inventor & Vault, Fusion 360, SolidWorks, CATIA V5, complex assemblies, technical drawings, design for manufacturing
Robotics & Automation	Robot end-effectors, gripper design, robot fixtures, pneumatic systems, cable routing, sensor/mechanical integration
Prototyping & Manufacturing	FDM/SLA 3D printing, silicone casting, mold design, mechanical testing, assembly, CNC manufacturing
Simulation & Engineering Tools	MATLAB/Simulink, finite element analysis basics, Python, C/C++
Product Development	Concept development, iterative prototyping, mechanical validation, design reviews, practical problem solving

LANGUAGES

- > **Portuguese** – Native
- > **English** – C1, TOEFL certificate
- > **German** – C1, Studienkolleg and B.Sc. Mechanical Engineering at TUM completed in German
- > **Spanish** – B1, TUM language course